

Discussion

What the Template Is, and Is Not I

The pipeline measures the *shape* of a literature — its size, growth, subfield composition, topical structure, citation geometry, and the hypotheses a field frames. It does not adjudicate scientific truth. The optional hypothesis scores summarize how the retrieved corpus *talks about* each claim, weighted by citation influence; they are an evidence-landscape instrument, not a verdict.

The 5 topics extracted by NMF provide a data-driven complement to the keyword-based subfield taxonomy. Where the taxonomy assigns each paper to a single bucket, the topics reveal overlapping thematic structure: a paper on modafinil's cognitive effects in ADHD patients belongs to the “Psychiatry” subfield but also loads on the “Cognitive Enhancement” and “ADHD Treatment” topics. This multi-resolution view is more informative than either approach alone.

Engine Coverage and Bias I

For this live run, the corpus is dominated by OpenAlex (1,000 records) and Crossref (1,000 records), with PubMed contributing 986 records and arXiv contributing 1. Semantic Scholar was rate-limited (HTTP 429) and returned zero records — a known limitation of its unauthenticated API tier. SovietRxiv and ChinaRxiv returned zero records for the modafinil query, which is expected given their coverage domains.

The max-results cap of 1,000 per engine means the full literature is larger than the retrieved corpus; the 2302 records represent a bounded sample rather than the complete literature. The citation network resolution rate of 22.6% reflects this: many cited works lie outside the retrieved slice. Increasing the cap or adding more engines would improve coverage but also increase runtime and API load.

Honest Defaults I

The committed seed corpus is synthetic (reserved test DOIs, generated authors) so that the whole pipeline runs offline and byte-identically. Its numbers demonstrate the machinery; they are not empirical findings about modafinil. Real claims require a live retrieval run with regenerated figures, reports, and manuscript variables — as produced in this instance.

Limitations and Extensions I

Several limitations bound the interpretation of results:

- ▶ **Coverage is bounded by the enabled engines and the query.** The max-results cap truncates each engine's contribution. Semantic Scholar's rate limiting excluded a major source; a Semantic Scholar API key would resolve this.
- ▶ **Subfield classification is keyword-based** and only as good as the configured taxonomy. Ambiguous papers may be misclassified; a classifier based on embeddings or supervised learning could improve accuracy.
- ▶ **The default embeddings are lexical** (TF-IDF/SVD). They capture term co-occurrence but not semantic similarity; a transformer backend (embeddings extra) would improve the quality of nearest-neighbour retrieval and clustering.

Limitations and Extensions II

- ▶ **Hypothesis scoring depends on an external language model.** Without Ollama running, scores read *pending*. The scoring is also sensitive to prompt design and model choice; the default `gemma3:4b` is a lightweight model suitable for demonstration but may miss nuanced assertions.
- ▶ **Abstract coverage is 55.5%.** Text analytics operate only on the subset of records with abstracts, biasing topic models and embeddings toward well-indexed sources.

Each limitation is a configuration or dependency choice rather than a change to the core architecture.